

Thow Min Jerald Cham

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ACADEMIC APPOINTMENTS

Nanyang Assistant Professor School of Electrical and Electronic Engineering, Nanyang Technological University	2026-Present
David and Ellen Lee Postdoctoral Scholar Division of Physics, Math and Astronomy, California Institute of Technology	2024-2026

EDUCATION

Cornell University , Ph.D., Physics Thesis Topic: Spin dynamics in antiferromagnetic van der Waals Heterostructures Advisor: Prof. Daniel C. Ralph Committee: Prof. Kin Fai Mak, Prof. Erich J. Mueller	2018-2024
Cornell University , M.S. Physics	2018-2021
Imperial College London , Physics BSc, 1 st class honors Research Advisor: Prof. John Tisch	2014-2017

AWARDS & FELLOWSHIPS

David and Ellen Lee Prize Postdoctoral Fellowship, California Institute of Technology	2024-2027
American Physical Society DMP Ovshinsky Award	2024
American Physical Society GMAG Student Travel Award	2024
A*STAR National Science Scholarship Roll of Honor	2017
Imperial College Ken Allen Prize (Top 5 students in Physics)	2017
Imperial College G Research Prize (Top 10 students in Physics, Computing and Math)	2015
Imperial College Physics Department Dean's List	2015, 2017
A*STAR National Science Scholarship	2014-2023

RESEARCH INTERESTS

Antiferromagnetic magnonics and spintronics

Generation, detection, and electrical or optical control of antiferromagnetic magnons and spin dynamics in quantum materials and van der Waals heterostructures.

Ultrafast probing of symmetry-guided quantum coupling pathways

Ultrafast optical spectroscopy of coupled spin, lattice, charge, and orbital degrees of freedom, including light-driven control of collective excitations and transient states.

Nanoscale and terahertz spectroscopy

Development and application of nanoscale optical and terahertz techniques to probe high-frequency excitations in thin crystals, heterostructures, and quantum devices.

CONFERENCE PRESENTATIONS

Invited Talk ' <i>Spin-filter tunneling detection of antiferromagnetic resonance with electrically-tunable damping</i> ', American Physical Society (APS) March Meeting, Denver CO	2026
Invited Talk ' <i>Spin-filter tunneling detection of antiferromagnetic resonance with electrically-tunable damping</i> ', IEEE Annual Conference on Magnetism and Magnetic Materials, Palm Beach FL	2025
Invited Talk ' <i>Exploring new spin torque phenomena in van der Waals topological and magnetic heterostructures</i> ', SPIE Optics + Photonics, San Diego CA	2025
Invited Talk ' <i>Spin-orbit torque in magnetic heterostructures: exchange interactions and ultrasensitive Sagnac magneto-optic interferometry</i> ', Symposium on Low-Dimensional Magnetic Quantum Materials, Materials Research Society Spring Meeting & Exhibit, Seattle WA	2024
Focus Session Invited Talk ' <i>Spin dynamics and exchange interactions from a van der Waals antiferromagnet</i> ', American Physical Society (APS) March Meeting, Minneapolis MN	2024
Invited Talk ' <i>Exchange bias between magnetic van der Waals materials</i> ', Cornell NanoScale Science and Technology Facility Annual Meeting, Ithaca NY	2023
Poster Presentation ' <i>Exchange bias between van der Waals materials: tilted magnetic states and field-free spin-orbit-torque switching</i> ', Spin Dynamics in Nanostructures Gordon Research Conference, Les Diablerets, Switzerland	2023
Symposium Invited Talk ' <i>Spin manipulation in van der Waals ferromagnets and antiferromagnets</i> ', Symposium on Magnetic Manipulation and Excitations in 2D Magnets, APS March Meeting, Las Vegas NV (on behalf of Prof. Dan Ralph).	2023
Contributed Talk ' <i>Anisotropic gigahertz frequency antiferromagnetic resonance in easy-axis van der Waals semiconductor</i> ', APS March Meeting, Las Vegas NV	2023
Poster Presentation ' <i>Anisotropic gigahertz frequency antiferromagnetic resonance in layered van der Waals semiconductor</i> ', Two Dimensional Electronics Beyond Graphene, Gordon Research Conference, Manchester NH	2022
Contributed Talk: ' <i>Anisotropic gigahertz frequency antiferromagnetic resonance in layered van der Waals semiconductor</i> ', APS March Meeting, Chicago IL	2022
Contributed Talk, ' <i>Spin-orbit torques generated by low-symmetry ZrTe₃</i> ', APS March Meeting	2021

REFEREED PUBLICATIONS

- 9 **Cham, T. M. J.**, Chica, D. G., Huang, X., K. Watanabe, T. Taniguchi, Roy, X., Luo, Y. K. and Ralph, D. C., "Spin-filter tunneling detection of antiferromagnetic resonance with electrically-tunable damping." *Science* 389, 479-482 (2025).
- 8 Huang, X., Song, Q., Gurung, G., Pharis, D. A., **Cham, T. M. J.**, Chen, Y., Jain, R., Olszewski, M., Feng, Y., El-Ghazaly, A., Tsymbal, E., Schlom, D. G., and & Ralph, D. C. "Turn-on of current-induced spin torque upon noncollinear antiferromagnetic ordering in delafossite PdCrO₂." *Advanced Materials*, e02670 (2025)
- 7 Zhou, W., Bishop, A.J., Zhang, X.S., Robinson, K., Lyalin, I., Li, Z., Bailey-Crandell, R., **Cham, T. M. J.**, Cheng, S., Luo, Y.K., Ralph, D.C., Muller, D. A., and Kawakami, R. K., "Tuning the Curie temperature of a 2D magnet/topological insulator heterostructure to above room temperature by epitaxial growth." *Physical Review Materials*, 7(10), 104004-104011 (2023).
- 6 **Cham, T. M. J.***, Dorrian, R., Zhang, X. S., Dismukes, A. H., Chica, D. G., Roy, X., May A.F., Ralph, D. C., & Luo, Y. K.*, "Exchange bias between van der Waals materials: tilted magnetic states and field-free spin-orbit-torque switching." *Advanced Materials*, 2305739 (2023).

- 5 Karimeddiny, S.*, **Cham, T. M. J.***, Smedley, O., Ralph, D.C. and Luo, Y.K.*, "Sagnac interferometry for high sensitivity optical measurements of spin-orbit torque." *Science Advances*, 9(36), eadi9039 (2023). First version: arXiv: 2109.13759.
- 4 **Cham, T. M. J.***, Karimeddiny, Zhang X. S., Dismukes, A. H., Roy, X., Ralph, D. C., & Luo, Y. K.*, "Anisotropic gigahertz antiferromagnetic resonances of the easy-axis van der Waals antiferromagnet CrSBr." *Nano Letters*, 22(16), 6716-6723 (2022).
- 3 **Cham, T. M. J.**, Karimeddiny, S., Gupta, V., Mittelstaedt, J. A., & Ralph, D. C. "Separation of artifacts from spin-torque ferromagnetic resonance measurements of spin-orbit torque for the low-symmetry van der Waals semi-metal ZrTe₃." *Advanced Quantum Technologies*, 5(2), 2100111 (2022).
- 2 Gupta, V., **Cham, T. M. J.**, Stiehl, G. M., Bose, A., Mittelstaedt, J. A., Kang, K., Jiang, S., Mak, K.F., Shan, J. & Ralph, D. C. "Manipulation of the van der Waals magnet Cr₂Ge₂Te₆ by spin-orbit torques." *Nano Letters*, 20(10), 7482-7488 (2020).
- 1 Tan, T. T., **Cham, T. M. J.**, Reithofer, M. R., Hor, T. A., & Chin, J. M. "Motorized Janus metal organic framework crystals." *Chemical Communications*, 50(96), 15175-15178 (2014).

*Equal contribution

MANUSCRIPTS IN PREPARATION

- 4 Yu, X., Berg, E., Khurana, B., Jain, R., **Cham, T. M. J.**, Smedley, O., Ross, C. A., Ralph, D. C. and Luo, Y. K., "Sagnac interferometry for high-sensitivity spin-orbit torque measurements in magneto-optical ferromagnetic insulator bismuth-substituted yttrium iron garnet." (2025)
- 3 Smedley, O., **Cham, T. M. J.**, Zheng, X. Y., Channa, S., Riddiford, L., Suzuki, Y., Luo, Y. K. & Ralph, D. C. "Sagnac Interferometer Measurement of Spin-Orbit-Torque Efficiency Acting on the Insulating Magnet Li_{0.5}(Al,Fe)_{2.5}O₄." (2025)
- 2 **Cham, T. M. J.**, Zhao, M.*, Zhou, W.*, Koerner, A., Le, D., Li, Z., Powalla, L., Bergner, D., Thareja, E., Alam, S., Wintz, S., Weigand, M., Hwang, J., Gayles, J., Ralph, D. C., Kawakami, R. K., Luo, Y.K., "Stabilizing bubble magnetic domains in epitaxial 2D magnet/topological insulator heterostructures through interfacial engineering." (2025)
- 1 Huang, X., Pharis, D. A., Zhou, H., Tian, Z., **Cham, T. M. J.**, Lee, K., Li, Y., Wang, C., Liang, Y., Olszewski, M., Yi, D., Eom, C., Schlom, D. G., Lane, W. M., Shao, D. & Ralph, D. C. "Control of out-of-plane anti-damping spin torque with a canted ferromagnetic spin source." (2025)

ACADEMIC SERVICE

Reviewer for Nature Communications, Newton, AIP Advances, Journal of Applied Physics, Applied Physics Letters, ACS Nano, Physical Review B.

Session chair for Moore Foundation Postdoctoral Symposium (2025), APS March Meeting (2023).

TEACHING, MENTORING & OUTREACH

Teaching Assistant/Grader for Cornell Physics 2216 Special Relativity course.	2019
Mentored undergrad Reiley Dorian for senior thesis, currently graduate student in Caltech Applied and Engineering Physics.	2022-2023
Mentored junior graduate student Orion Smedley, at Cornell Physics.	2022-2024
Mentoring three junior graduate students and two undergraduate students from Luo group at the University of Southern California Physics department, as part of collaborative project involving Scanning Transmission X-ray Magnetic Circular Dichroism (ST-XMCD) measurements of nano-scale magnetic domains in vdW heterostructures using the Maxymus end-station at BESSY II	2024-2026

beamline, Helmholtz Zentrum Berlin, Germany.

Mentored high school students as part of summer research at Hsieh group, Caltech 2026

Singapore National Science Challenge organizing committee member and judge. 2017

Guest science teacher for Windsor Central High School New York, with Cornell Center for Materials Research (CCMR). Guided students through “Build your own speaker” and “Red cabbage pH meter” lab activity. 2023

Designed circuits activity for local middle school outreach with Cornell Center for Materials Research (CCMR). Developed class materials and kits for demonstrating the efficiency of LED bulbs in comparison with incandescent bulbs. 2023

Lecture on “Introduction to two-magnon scattering in Raman spectroscopy” and hosted lab tours for students from Pasadena High School. 2025